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Diagrams, charts, tables

Part 2: Item designation

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STANDARDS AND CERTIFICATION
NATIONAL BUREAU OF STANDARDS



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

DIAGRAMS, CHARTS, TABLES

Part 2: Item designation

FOREWORD

- 1) The formal decisions or agreements of the I E C on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote this international unification, the I E C expresses the wish that all National Committees having as yet no national rules, when preparing such rules, should use the I E C recommendations as the fundamental basis for these rules in so far as national conditions will permit.
- 4) The desirability is recognized of extending international agreement on these matters through an endeavour to harmonize national standardization rules with these recommendations in so far as national conditions will permit. The National Committees pledge their influence towards that end.

PREFACE

This Recommendation has been prepared by Sub-Committee SC 3B, Preparation of Diagrams, Charts and Tables. Item Designation, of I E C Technical Committee No. 3, Graphical Symbols.

A draft, prepared by the Secretariat of SC 3B, was discussed at the meeting held in Teheran in 1969, as a result of which a new draft was submitted to the National Committees for approval under the Six Months Rule in February 1970.

The following countries voted explicitly in favour of publication of Part 2:

Australia	Germany
Austria	Israel
Belgium	Italy
Canada	Japan
Czechoslovakia	Netherlands
Denmark	Switzerland
Finland	Turkey
France	United States of America

DIAGRAMS, CHARTS, TABLES

Part 2 : Item designation

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STANDARDS AND SAFETY SECTION
NATIONAL ENERGY AUTHORITY

1. Scope

This Recommendation covers rules for the unambiguous formation and application of discrete item designations for electrical parts and equipment.

The designation is shown at an appropriate place near the graphical symbol of the item. The designation correlates the item on different diagrams, parts lists, circuit descriptions, instructions and in the equipment. For maintenance purposes, the designation or part of it may also be shown on or near the item in the equipment.

2. Definitions

For the purpose of this Recommendation, the following definitions shall apply.

2.1 Designation

A distinctive code, which serves to identify an item in a diagram, list, chart and on the equipment.

2.2 Diagram

A diagram may show the manner in which the various parts of a network, installation, group of apparatus or items of an apparatus are interrelated and/or interconnected.

2.3 Chart

A chart (U.S.A.: graph) may show the interrelation between:

- a) different operations;
- b) operations and time;
- c) operations and physical quantities;
- d) the state of several items.

2.4 Table

A table replaces or supplements a diagram or a chart.

2.5 Block of designation

Related information is grouped in a single designation block. The type and amount of information given by a designation will depend on the type of diagram.

2.6 Item

Used for component, equipment, plant, unit, etc., which is represented by a graphical symbol on a diagram.

2.7 Kind of item

The code for the kind of item is derived from sort, variety, class or group of items regardless of their function in a circuit.

2.8 *Function*

Characteristic action or purpose of an item in relation to others. The designation of the function can be rather general, e.g. a relay may have an auxiliary function (auxiliary relay). The designation of the function can also be more specific, e.g. a motor may have the function of driving a pump of the cooling system of a generator.

2.9 *Number*

A number should be assigned to each item in an unambiguous manner. The numbers need not necessarily be in an uninterrupted sequence. Groups of numbers may be assigned to groups of items if desired.

2.10 *Location*

The physical position of an item in a subassembly, unit, plant, etc. The location designation may be essential to identify the item, e.g. for maintenance.

2.11 *Higher-level assignment*

A supplementary designation may be assigned to an item, if it is desired to express its relation to a larger unit of a system, of which it is a part. The designation may be derived from the kind, purpose or location of the larger unit.

2.12 *Qualifying symbols*

For example hyphen, colon, etc. are used to identify the various designation blocks.

3. **Purpose of different types of designation**

An item designation may be used for general or special purposes depending on the kind of information required. The different types of designation dealt with in this Recommendation provide information for the following purposes:

- 1) Higher-level assignment showing correlation with other parts of the equipment with regard to location and/or function.
- 2) Location of item.
- 3) Identification of item:
 - A = kind of item;
 - B = number of item;
 - C = function of item.
- 4) Terminal and conductor marking.

On most diagrams an appropriate portion of the item designation (for the purposes listed above) is sufficient. The selection of the portion used depends on the type of diagram.

4. **Arrangement of item designation**

- 4.1 The kind, number and function of an item, and sometimes its higher-level assignment, are known at an early stage of the design and development of the equipment. Therefore, these portions of the designation can be used on sketches and diagrams.

Qualifying symbols are used to distinguish the various blocks 1, 2, 3 and 4 of a complete designation. The different blocks with their qualifying symbols are shown below:

Block 1 =

HIGHER- LEVEL ASSIGN- MENT

Block 2 +

LOCATION

Block 3 —

3A	3B	3C
KIND	NUMBER	FUNCTION

Block 4 :

TERMINAL

If qualifying symbols other than those shown above are used, they must be explained.

The qualifying symbol may be omitted or replaced by a note on the diagram if there is no ambiguity. Where the designations appear in lists (e.g. connection tables, parts lists) the columns of the list may be so arranged that the qualifying symbols can be omitted.

The following sequence of designation blocks is preferred:

=

1
HIGHER- LEVEL ASSIGN- MENT

 +

2
LOCATION

 —

3A	3B	3C
KIND	NUMBER	FUNCTION

 :

4
TERMINAL

Where there is a lack of space on the diagram or when only some of the designation blocks are required, their sequence or presentation may be varied.

It is recommended that the same sequence be used for all designations in a particular set of diagrams.

- 4.2 In this Recommendation, the designations consist of latin letters and arabic numbers. Their mutual arrangement is defined in Clause 5.

To make possible the use of automatic data processing and to simplify the use of designations for preparation of lists, wiring instructions, etc. lower case and upper case letters should have the same meaning. Where this is not possible (e.g. for terminal designations), means shall be provided to make the distinction.

5. Designation groups

5.1 Initial remarks

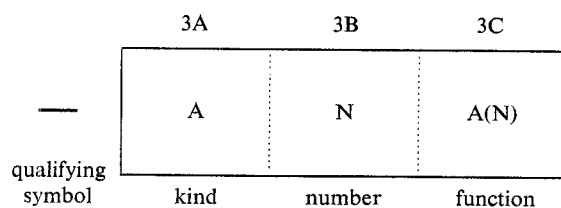
The variety of elements and equipment in electrotechnology and their rapid development make it impracticable to establish all necessary designations in a standard.

By grouping several related items, however, it has been possible to create recommended lists of designations (see Tables I and II, pages 19 and 21). For cases where these lists are not applicable, this Recommendation provides general rules for the establishment of the designations in each of the blocks defined in Clause 4. The meaning of designations built up according to these rules shall be explained or referenced on the diagram or explained in supporting documentation.

As designations consist of letters and/or numbers, it has been necessary in certain cases to give mandatory rules for the sequence of letters and numbers. In the following rules, A stands for letter(s) and N stands for number(s).

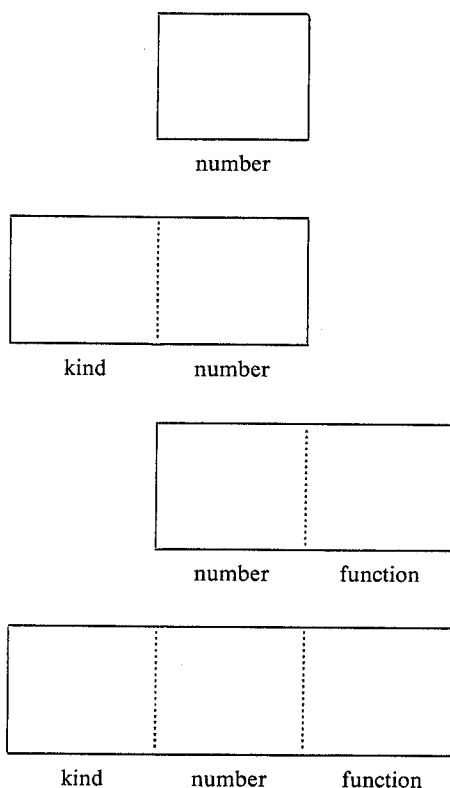
In Sub-clause 4.1, a preferred sequence of the four blocks of designation is recommended. However, in the following the blocks are dealt with in their order of importance.

5.2 Identification and function of item



This block consists of three parts (3A, 3B and 3C). Part 3B is mandatory. Parts 3A or 3C or both may supplement 3B.

The following are possible combinations:



- 5.2.1 Part 3A gives information about the *kind* of item or element. This should consist of one or more letters, preferably one.

Letters for designating the kind of item are often derived mnemonically from a language. For international use, however, it is useful to have a fixed list of letters. Because the graphical symbol on a diagram provides additional information, it is possible to allot a single letter to each group of items.

The recommended list of kind of item code letters is shown in Table I, page 19. If another list is used, it shall be shown or referenced on the diagram.

- 5.2.2 Part 3B consists of a *number* which serves:

- a) to identify the item without using either part 3A or part 3C of this block;
- b) to distinguish between several items designated by the same letter in part 3A or 3C;
- c) to indicate whether the letter used is designating the kind of item (letter followed by number) or the function (letter preceded by number).

Sequential numbers from 1 to *n* may be used. Groups of numbers may be assigned to groups of items.

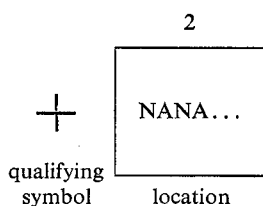
An additional number, separated by a point (.), may be used to distinguish between similar parts of an item, which are shown separately on a drawing, e.g. the contact units of a relay shown in detached representation.

- 5.2.3 Part 3C gives information about the *function* of an item. When using this designation to describe the special function of an item in a circuit, the variety of functions prohibits the creation of a complete code.

The designation can therefore be chosen arbitrarily, but it shall begin with a letter, which may be followed by additional letters and/or numbers if necessary. The coding shall be explained.

Table II, page 21, provides a list of letters for use where only general information about the function is sufficient (e.g., main, auxiliary, etc.).

5.3 Location

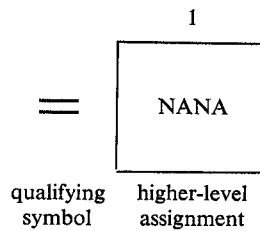


The location method permits rapid physical location of items in large, complicated equipment featuring multiple use of many identical, or closely similar, items. Either of two methods of assigning numbers, sequential or coordinate, may be used at any equipment level, based on equipment configuration. These methods shall be applied in such a way that duplicate complete item designations do not occur in a single equipment or system.

A location designation generally consists of an alternation of letters and numbers which describe the coordinates (for example, see Figure 1, page 22). However, in certain cases, location designations consisting of numbers only may be used (for example, see Figure 2, page 22). In either case, the associated drawings will illustrate the code assignments (for example, see Figure 3, page 23).

If other location systems are used, they shall be explained.

5.4 Higher-level assignment

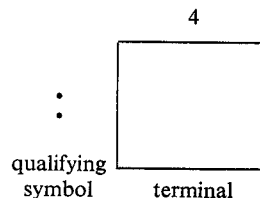


Generally, each unit shall be assigned an identifying number, beginning with 1 and running consecutively. This number is the item designation of the unit. If there is only one unit, the unit number is omitted.

Different unit numbers shall be used to identify identical or similar units of a system. However, item designations for assemblies and parts within these units shall be the same for corresponding items.

If necessary, a unit location designation may be developed and used to prefix the unit number. The location designation required for any particular application depends on the equipment configuration.

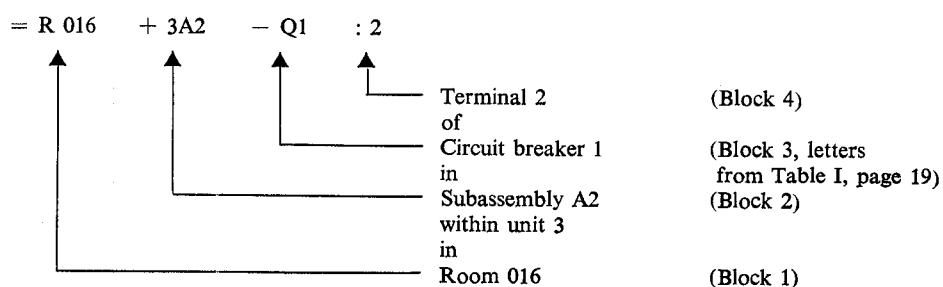
5.5 Terminals and conductors



Terminal designations shall correspond to the markings on the item. When terminal and conductor markings have been standardized in Recommendations of the relevant Technical Committee of the I E C, those markings shall be used. When terminals and conductors are not marked and no Recommendation or national standards for such marking exist, arbitrary designation should be assigned on the diagram. These designations are used, if necessary, as part of the complete item designation.

6. Examples

a) Complete item designation in which the higher-level assignment is used to indicate the *location* of the item in the complete plant:



b) Item designation in which the higher-level assignment is used to indicate the *purpose* of the item in the complete plant:

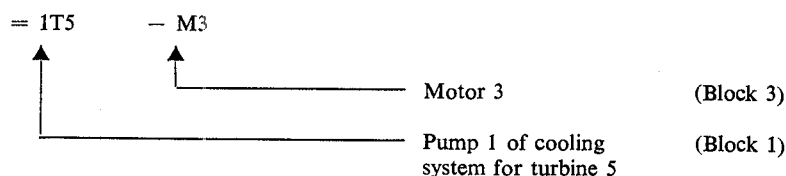


TABLE I

Letter codes for the designation of kind of item (Complying to block 3A)

Letter code	Kind of item	Examples
A	Assemblies, subassemblies	Amplifier with tubes or transistors, magnetic amplifier, laser, maser
B	Transducers, from non-electrical quantity to electrical quantity or vice-versa	Thermo-electric sensor, thermo cell, photo-electric cell, dynamometer, crystal transducer, microphone, pick-up, loudspeaker, synchros, resolvers
C	Capacitors	
D	Binary elements, delay devices, storage devices	Combinative elements, delay lines, bistable elements, monostable elements, core storage, register, magnetic tape recorder, disk recorder
E	Miscellaneous	Lighting devices, heating devices, devices not specified elsewhere in this Table
F	Protective devices	Fuse, over-voltage discharge device, arrester
G	Generators, supplies	Rotating generator, rotating frequency converter, battery, supply device, oscillator, quartz-oscillator
H	Signalling devices	Optical and acoustical indicators
J		
K	Relays, contactors	
L	Inductors	Induction coil, line trap
M	Motors	
N		
P	Measuring equipment, testing equipment	Indicating, recording and integrating measuring devices, signal generator, clocks
Q	Mechanical switching devices for power circuits	Circuit-breaker, isolator
R	Resistors	Adjustable resistor, potentiometer, rheostat, shunt, thermistor
S	Switches, selectors	Control switch, push-buttons, limit switch, selector switch, selector, dial contact, connecting stage
T	Transformers	Voltage transformer, current transformer
U	Modulators, changers	Discriminator, demodulator, frequency changer, coder, inverter, converter, telegraph translator
V	Tubes, semiconductors	Electronic tube, gas-discharge tube, diode, transistor, thyristor
W	Transmission paths, waveguides, aerials	Jumper wire, cable, busbar, waveguide, waveguide directional coupler, dipole, parabolic aerial
X	Terminals, plugs, sockets	Disconnecting plug and socket, test jack, terminal board, soldering terminal strip
Y	Electrically operated mechanical devices	Brake, clutch, pneumatic valve
Z	Terminations, hybrid transformers, filters, equalizers, limiters	Cable balancing network, compandor, crystal filter

TABLE II

Letter codes for the designation of general function (Complying to block 3C)

Letter code	General function
A	Auxiliary
B	Direction of movement (forward, backward, hoist, lower, clockwise, anti-clockwise)
C	Counting
D	Differentiating
E	
F	Protecting
G	Testing
H	Signalling
J	Integrating
K	Jog
L	
M	Main
N	Measuring
P	Proportional
Q	State (start, stop, limit)
R	Reset, erase
S	Storing, recording
T	Timing, delaying
U	
V	Speed (accelerating, braking)
W	Adding
X	Multiplying
Y	Analogue
Z	Digital

D 7
 Repère vertical Repère gauche à droite
 Vertical location Left-to-right location

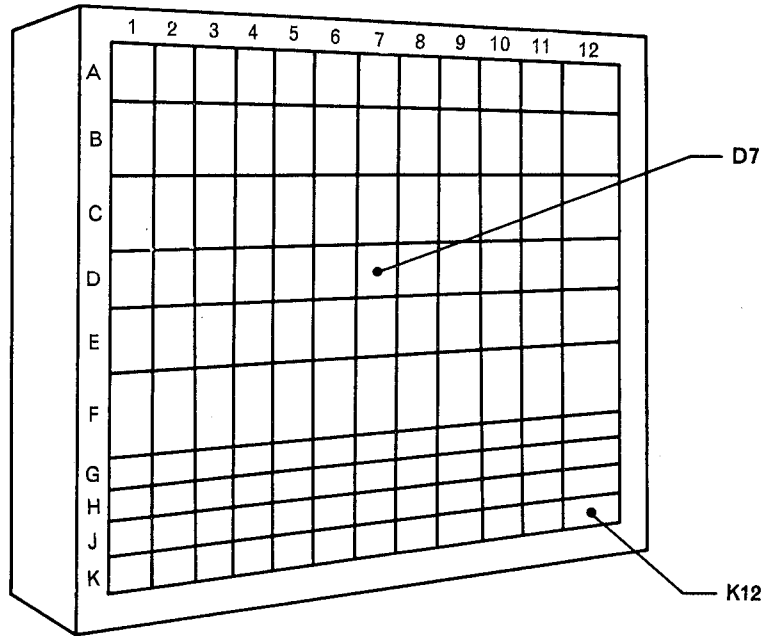


FIG. 1. — Repérage d'emplacement alpha-numérique.
Alpha-numeric location.

4 07
 Repère vertical Repère gauche à droite
 Vertical location Left-to-right location

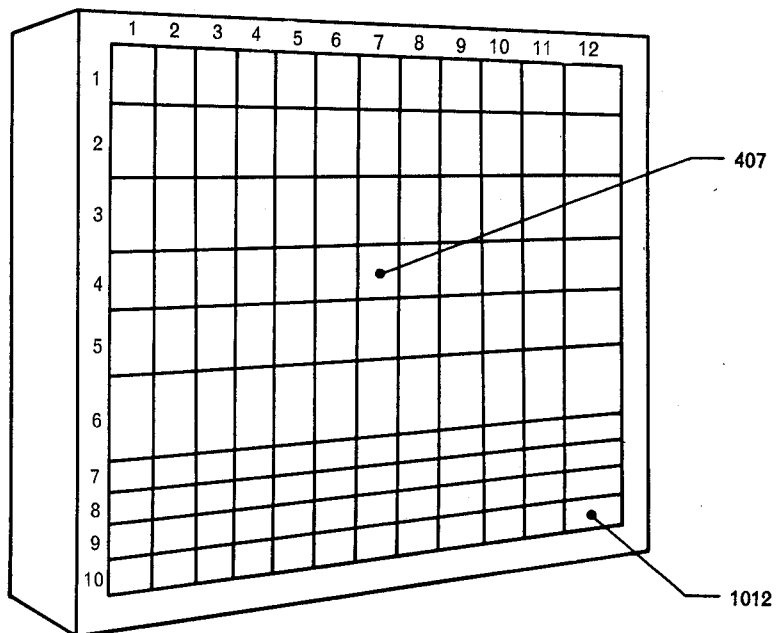


FIG. 2. — Repérage d'emplacement numérique.
Numeric location.

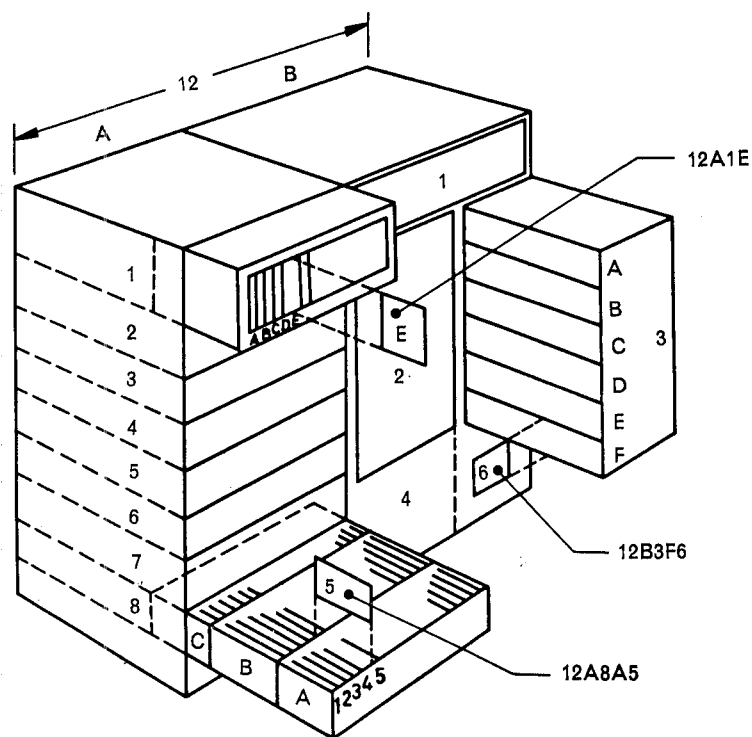


FIG. 3. — Illustration d'un code de repérage de l'emplacement.
Example of a location designation code.